# Practice workbook · Match digestive enzymes to substrate, product and site

Amylase, maltase, proteases and lipase: their substrates, products and where they are made or act.

12 question parts: 1 worked-example retry and 11 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: HBI-DIGESTIVE-ENZYME-MATCH

# 1. Question 2 · part a.i · 3 marks

June 2023 Paper 2R

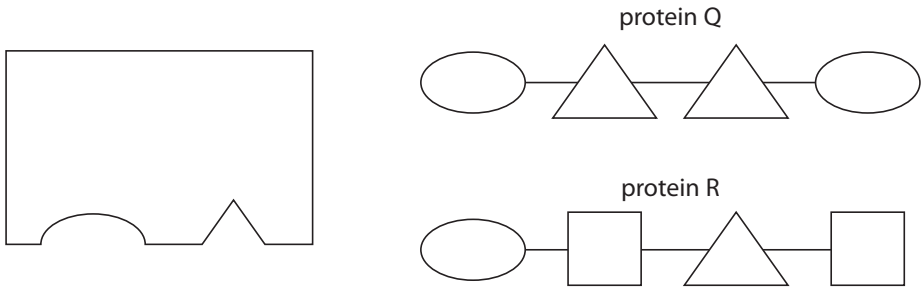
Worked-example retry · Attempt part a.i only.

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**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | | |
|---|---|---|---|---|
| amino acids | amylases | fatty acids | proteases | |
| protein Q | protein R | starch | sugar | |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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## 1. Question 2 · part a.i · 3 marks · Working and answer

Working and answer

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## 2. Question 3 · part b · 1 mark

June 2019 Paper 2

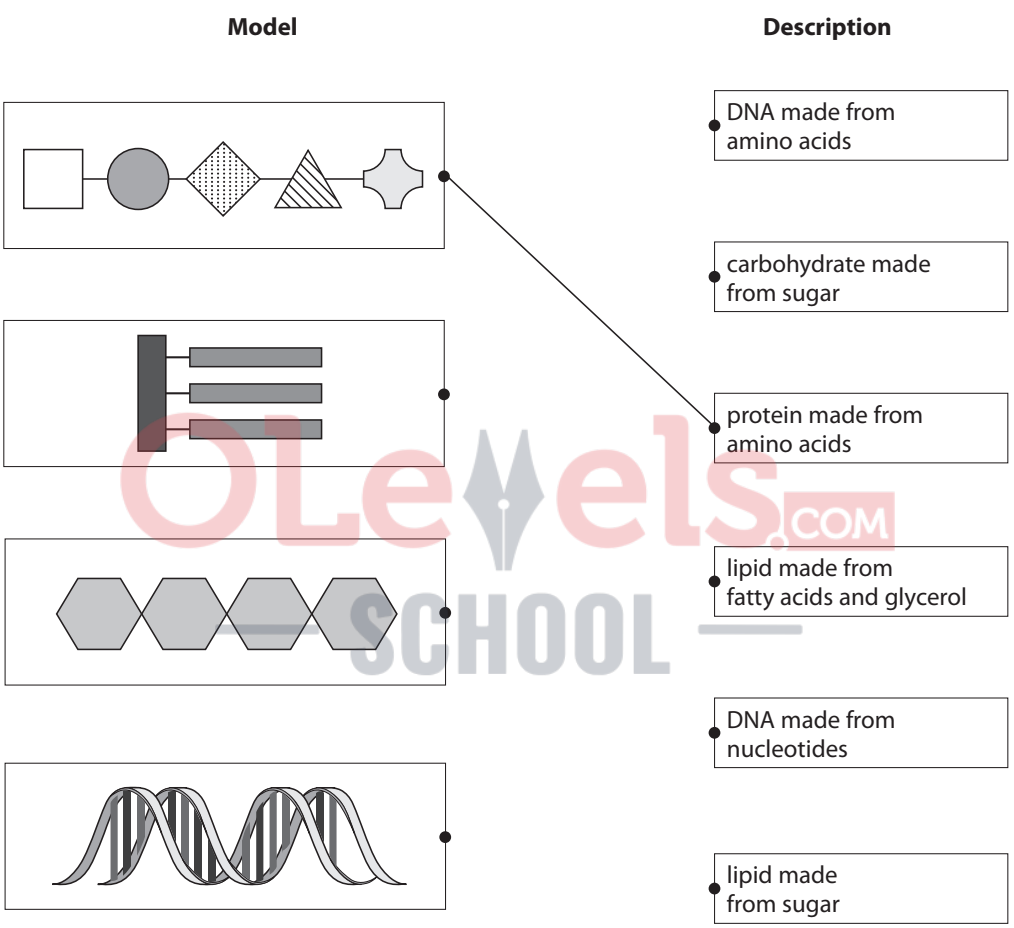
Further practice · Attempt part b only.

**3** (a) The diagram shows models of four biological molecules.

Draw a straight line from each model to its correct description.

One has been done for you.

**(3)**




(b) Which body organ produces enzymes that break down protein?

**(1)**

- ☒ **A** gall bladder
- ☒ **B** large intestine
- ☒ **C** mouth
- ☒ **D** stomach

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P 5 8 5 6 6 A 0 6 2 8

## 2. Question 3 · part b · 1 mark

June 2019 Paper 2

Further practice · Attempt part b only.

(c) A student uses this method to test different substances for protein.

- grind solid substances into small pieces
- place substances into separate test tubes
- add a few drops of Biuret reagent to each substance
- record the colour change for each substance

The table shows the student's results.

| Substance | Colour after Biuret test |
|---|---|
| milk | purple |
| pasta | blue |
| lemon juice | blue |
| cheese | purple |
| distilled water | purple |

(i) In the student's test, how many of the substances give a positive result for protein?

**(1)**

.....

(ii) The result for distilled water is incorrect.

Suggest one mistake the student could have made to get this incorrect result.

**(1)**

.....

.....

(d) The table shows an incomplete risk assessment for the Biuret test.

Complete the table by describing how to reduce the risk of each hazard.

**(2)**

| Hazard | Reducing risk |
|---|---|
| stools – trip hazard | keep stools under bench |
| broken glass – cuts | |
| Biuret reagent – irritant | |

**(Total for Question 3 = 8 marks)**

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## 2. Question 3 · part b · 1 mark · Working and answer

Working and answer

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## 3. Question 1 · part b.i · 1 mark

November 2020 Paper 2R

Further practice · Attempt part b.i only.

**1** Food enters the alimentary canal through the mouth.

All undigested food leaves the body through the anus.

(a) Name three structures the food passes through between the mouth and the anus.

**(3)**

1 .....

2 .....

3 .....

(b) Some of the food is digested by enzymes as it passes through the alimentary canal.

(i) Name an enzyme that digests protein.

**(1)**

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(ii) Describe the processes that occur in the mouth to digest food.

**(4)**

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2

P 6 2 0 5 4 A 0 2 2 4

## 3. Question 1 · part b.i · 1 mark · Working and answer

Working and answer

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# 4. Question 2 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

**2** A student investigates the effect of bile salts on the digestion of lipids. The student uses milk as a source of lipids.

The student sets up three test tubes.

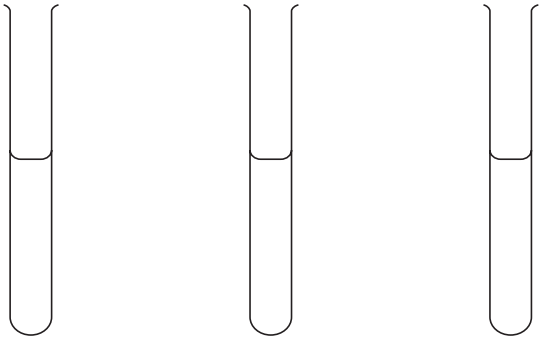



Table 1 lists the contents of each test tube.

| Test tube A | Test tube B | Test tube C |
|---|---|---|
| 5 $cm^3$ of milk | 5 $cm^3$ of milk | 5 $cm^3$ of milk |
| 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate | 2 $cm^3$ of sodium hydrogencarbonate |
| 6 drops of phenolphthalein | 6 drops of phenolphthalein | 6 drops of phenolphthalein |
| 1 $cm^3$ enzyme | 1 $cm^3$ enzyme | 1 $cm^3$ boiled enzyme |
| distilled water | bile salts | bile salts |

**Table 1**

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# 4. Question 2 · part a.i · 1 mark

November 2020 Paper 2

Further practice · Attempt part a.i only.

The student records the colour of the contents of each tube at the start of the investigation, and at 5 minute intervals for 15 minutes.

Table 2 shows the student's results.

| | Colour of contents | | |
|---|---|---|---|
| | **Tube A** | **Tube B** | **Tube C** |
| at start | pink | pink | pink |
| after 5 minutes | pink | colourless | pink |
| after 10 minutes | pink | colourless | pink |
| after 15 minutes | colourless | colourless | pink |

**Table 2**

Phenolphthalein is pink in solutions above pH 10 and colourless in solutions below pH 8.

(a) (i) Which enzyme is used in this investigation?

**(1)**

- ☒ **A** amylase
- ☒ **B** carbohydrase
- ☒ **C** lipase
- ☒ **D** protease

(ii) Which chemical elements are found in lipids?

**(1)**

- ☒ **A** carbon, hydrogen and oxygen
- ☒ **B** carbon, hydrogen, oxygen and nitrogen
- ☒ **C** carbon, hydrogen, oxygen and sulfur
- ☒ **D** carbon, hydrogen, oxygen, nitrogen and sulfur

(iii) Where in the body are bile salts produced?

**(1)**

- ☒ **A** brain
- ☒ **B** duodenum
- ☒ **C** liver
- ☒ **D** pancreas

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## 4. Question 2 · part a.i · 1 mark · Working and answer

Working and answer

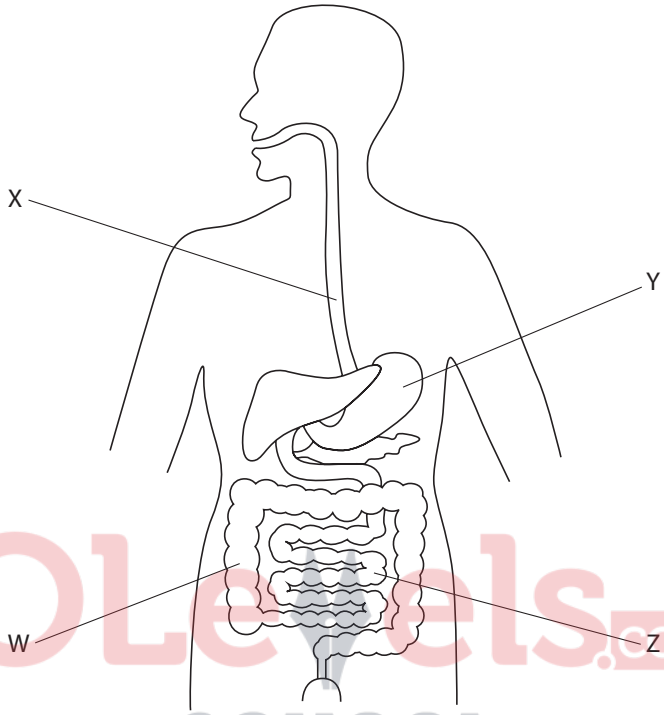
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# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

**2** (a) The diagram shows the human digestive system.




(i) Name the structures labelled W, X, Y and Z.

**(4)**

W .....

X .....

Y .....

Z .....

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P 7 0 9 4 9 A 0 6 2 4

# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

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(ii) The box gives words used to describe how food is moved through structure X towards structure Y.

| | | | |
|---|---|---|---|
| relax | coordination | contract | liquid |
| peristalsis | muscles | enzymes | bolus |

Use words from the box to complete the passage.

**(4)**

Food forms a ..... at the back of the mouth.

The food is swallowed and passes into structure X. The walls of structure X contain .....

which ..... to push the food towards structure Y.

This process is known as ..... .

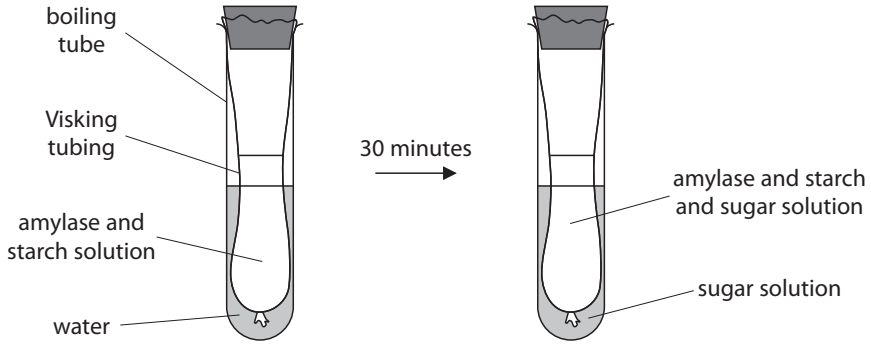
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# 5. Question 2 · part b.i · 1 mark

June 2022 Paper 1R

Further practice · Attempt part b.i only.

(b) The diagram shows some apparatus used to investigate the effect of an enzyme called amylase on starch.




(i) Name one area in the digestion system where amylase is produced.

(1)

(ii) Name the structure in the digestive system that the Visking tubing represents.

(1)

(iii) Explain the results of the investigation after 30 minutes.

(2)

(iv) Name the chemical used to test for sugar in the water surrounding the Visking tubing after 30 minutes.

(1)

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## 5. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

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# 6. Question 4 · part a.ii · 1 mark

June 2022 Paper 1R

Further practice · Attempt part a.ii only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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## 6. Question 4 · part a.ii · 1 mark · Working and answer

Working and answer

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## 7. Question 4 · part a.iii · 2 marks

June 2022 Paper 1R

Further practice · Attempt part a.iii only.

**4** (a) Lipids are digested in the small intestine.

(i) Describe a test for lipids in food.

**(3)**

(ii) Name the enzyme that digests lipids.

**(1)**

(iii) Name the products of lipid digestion.

**(2)**

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(b) A student investigates how the pH of a lipid solution, containing an enzyme that digests lipids, changes over 10 minutes at two different temperatures.

The tables show the student's results.

**Temperature at 37 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 8.5 | 8.0 | 7.5 | 7.2 | 7.0 | 6.8 | 6.2 | 5.8 | 5.5 | 5.5 |

**Temperature at 80 °C**

| Time in minutes | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| pH | 9.2 | 9.0 | 8.8 | 8.8 | | 8.6 | 8.6 | 8.4 | 8.4 | 8.4 |

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## 7. Question 4 · part a.iii · 2 marks · Working and answer

Working and answer

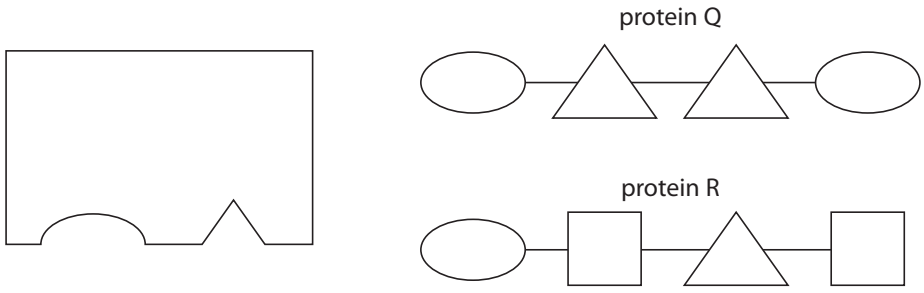
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# 8. Question 2 · part a.ii · 1 mark

June 2023 Paper 2R

Further practice · Attempt part a.ii only.

**2** (a) The diagram represents an enzyme, known as enzyme P, and two named substrates.




**Enzyme P** **Substrates**

(i) Use the diagram and words from the box to complete the passage about enzyme P.

**(3)**

| | | | |
|---|---|---|---|
| amino acids | amylases | fatty acids | proteases |
| protein Q | protein R | starch | sugar |

Enzyme P belongs to a group of enzymes called ..... .

This enzyme breaks down ..... in food

to ..... .

(ii) Give one location where enzyme P is found in the digestive system.

**(1)**

.....

(iii) Name the molecule that codes for the production of enzyme P.

**(1)**

.....

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## 8. Question 2 · part a.ii · 1 mark · Working and answer

Working and answer

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# 9. Question 4 · part a.vi · 1 mark

June 2024 Paper 1R

Further practice · Attempt part a.vi only.

**4** (a) The table gives information on how the time taken to digest starch by an enzyme varies at different pH levels.

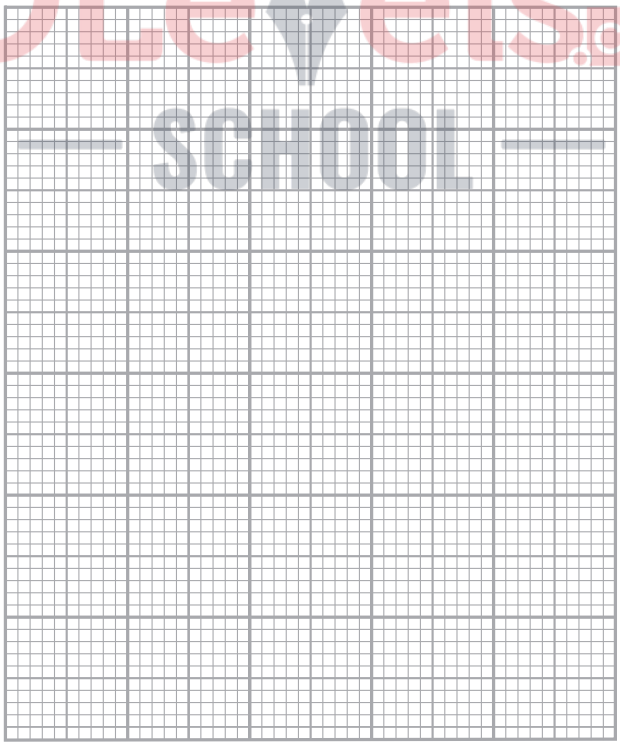
| pH | Time taken to digest starch in minutes |
|---|---|
| 2 | 60 |
| 4 | 45 |
| 6 | 20 |
| 8 | 35 |
| 10 | 55 |

(i) Plot a graph using the data in the table.

**(3)**

(ii) Draw a curve of best fit.

**(1)**



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## 9. Question 4 · part a.vi · 1 mark

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Further practice · Attempt part a.vi only.

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(iii) Use the graph to determine the optimum pH for the digestion of starch.

**(1)**

(iv) Describe the trend shown by the graph.

**(2)**

(v) Explain why the time taken to digest starch is different at pH 2 compared with the time taken at pH 6.

**(3)**

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(vi) Name a part of the body where starch digestion takes place.

**(1)**

(b) Describe a test for starch in a sample of food.

**(2)**

**(Total for Question 4 = 13 marks)**

## 9. Question 4 · part a.vi · 1 mark · Working and answer

Working and answer

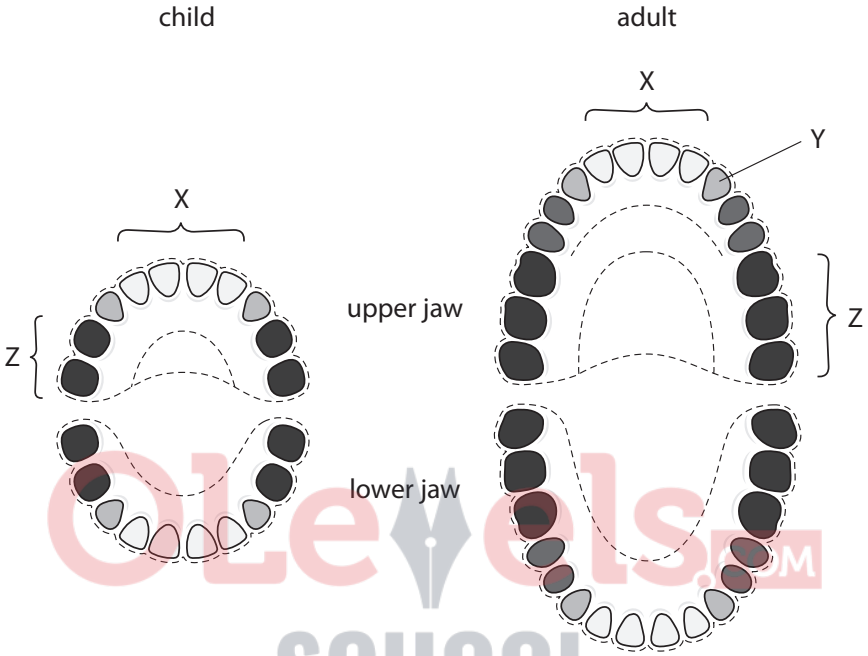
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# 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

**1** (a) The diagrams show the arrangement of teeth in a young child and the arrangement of teeth in an adult.




(Source: https://www.medicalnewstoday.com/articles/326754#number-of-teeth)

(i) Name the types of teeth shown by X and Z.

**(2)**

X .....

Z .....

(ii) State the function of the tooth labelled Y.

**(1)**

.....

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## 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

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(b) Digestion of food begins in the mouth.

Use the most appropriate words or numbers to complete the passage about digestion of food in the mouth.

**(4)**

Saliva in the mouth contains an enzyme called ..... .

This enzyme speeds up the digestion of starch to produce

..... .

The enzyme in saliva works best at pH ..... .

If the pH in the mouth was too low or too high, the digestion of carbohydrates

would ..... .

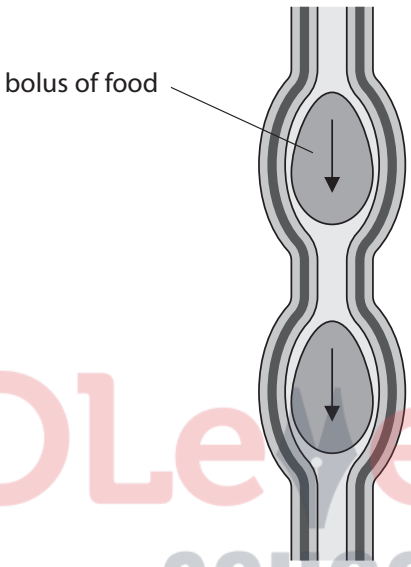
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## 10. Question 1 · part b · 4 marks

June 2024 Paper 2R

Further practice · Attempt part b only.

(c) The diagram shows the part of the digestive system that transports food from the mouth to the stomach.




(Source: https://www.motifolio.com/1031265.html)

(i) Name the part of the digestive system that transports food from the mouth to the stomach.

**(1)**

.....

(ii) Describe how the action of this part of the digestive system transports food from the mouth to the stomach.

**(2)**

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**(Total for Question 1 = 10 marks)**

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## 10. Question 1 · part b · 4 marks · Working and answer

Working and answer

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## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

**7** Six students investigate the activity of an enzyme found in the alimentary canal.

This enzyme digests a cloudy suspension into a clear solution.

The results of the investigation are shown in the table.

| Student | Volume of enzyme solution in $cm^3$ | Temperature in °C | pH | Volume of cloudy suspension in $cm^3$ | Time tubes left in minutes | Appearance at end of investigation |
|---|---|---|---|---|---|---|
| 1 | 5 | 37 | 2 | 10 | 9 | cloudy |
| 2 | 5 | 37 | 2 | 10 | 20 | clear |
| 3 | 5 | 37 | 4 | 10 | 35 | clear |
| 4 | 5 | 37 | 4 | 10 | 20 | cloudy |
| 5 | 5 | 37 | 6 | 10 | 8 | cloudy |
| 6 | 5 | 37 | 6 | 10 | 40 | cloudy |

(a) Explain why the investigation was carried out at 37 °C.

**(2)**

(b) Explain which property of the enzyme the students are investigating.

**(2)**

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## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

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(c) Comment on the results of the investigation.

In your answer include ideas of validity and accuracy.

**(4)**

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(d) Describe a control experiment that the students could have carried out.

**(3)**

## 11. Question 7 · part f · 2 marks

November 2024 Paper 2

Further practice · Attempt part f only.

(e) State a possible conclusion that the students could make from the results of this investigation.

**(2)**

(f) Explain where this enzyme is likely to be found in the alimentary canal.

**(2)**

**(Total for Question 7 = 15 marks)**

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Working and answer

## 12. Question 7 · part a · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a only.

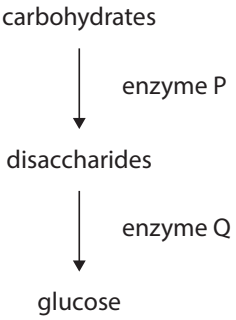
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**7** (a) The diagram shows how carbohydrates are broken down to glucose by two different enzymes, enzyme P and enzyme Q.

These enzymes are found in different regions of the digestive system.




Name enzyme P.

**(1)**

.....

(b) People with diabetes can use glucose testing strips to test their urine.

Describe how enzymes are used in glucose testing strips.

**(2)**

.....

.....

.....

.....

(c) People with type 1 diabetes are unable to produce hormones to control their blood glucose levels.

Name the organ responsible for producing hormones to control levels of glucose in the blood.

**(1)**

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P 7 4 6 3 2 A 0 1 9 2 4

## 12. Question 7 · part a · 1 mark · Working and answer

Working and answer

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